EDWARD BAKER

919-670-906 edwardunc1@gmail.com Github LinkedIn 💻 https://edwarddbaker.org

PROFESSIONAL SUMMARY

Machine learning professional with experience building and deploying production-grade models in cloud environments, leading cross-functional projects, and translating complex technical work for non-technical stakeholders. Strong background in applied analytics and software engineering across financial services, public sector, and enterprise data platforms.

Languages & Tools: Python, R, SQL, MATLAB, Java, C, HTML, JavaScript, JupyterHub, Git, Azure, AWS
Certifications: AWS Certified Cloud Practitioner

EDUCATION

M.S. in Analytics May 2024
Institute for Advanced Analytics, NC State University, Raleigh, NC

B.S. in Computer Science, Minor in Religious Studies May 2023
University of North Carolina at Chapel Hill, Chapel Hill, NC

EXPERIENCE & PROJECTS

Elevate
Data Scientist I June 2024—November 2025
Data Scientist II November 2025—Current

Tech Stack: Python, Azure DevOps, Azure ML SDK, Snowflake, SQL, Linux

- Engineered an end-to-end cloud pipeline in Azure ML to automate monthly reparameterization of **eight production credit-risk models**, reducing model refresh time from **two weeks to one day**
- Developed and productionized a high-performance credit-risk model using XGBoost, improving AUC by **13%**, and designed a modular data transformation layer (~**1,200** lines of SQL) to convert raw XML data in Snowflake into validated, consumable datasets
- Designed and deployed **containerized model inference APIs** using Docker and Azure Kubernetes Service (AKS), incorporating unit testing and standardized deployment workflows for production reliability.
- Designed, versioned, and maintained an internal Python package distribution system providing **50+ reusable modules** for feature engineering, validation, and model lifecycle management; adopted by a **15-member team** and integrated into **12+ production projects**
- Built parameterized, reproducible training and deployment pipelines in Azure ML to replace underperforming third-party vendor direct mail response models
- Created and led a structured onboarding program for **five new team members**, accelerating ramp-up and ensuring consistent engineering practices
- Elected lead for the NC State Institute for Advanced Analytics practicum project (Class of 2026), coordinating technical design, implementation, and stakeholder communication

U.S. Department of Education September 2023—April 2024
Team Leader

- Served as elected project lead for a master's capstone, providing technical guidance and project planning for a team of **5 students**
- Implemented custom network clustering algorithm based on **KNN** techniques to construct social networks by connecting individuals across **30+ attributes for 300K+ individuals**
- Delivered well-documented and reproducible Python code to client to create new network data for 45M+ customers

- Estimated the effect of one's social network on **5 long-term student outcomes** using logistic regression, decision trees, and chi-square tests

UNC Chapel Hill Computer Science Department Chapel Hill, NC
Undergraduate Learning Assistant – COMP301 Foundations of Programming August 2022—May 2023

- Pioneered a new assignment and facilitated a workshop for 50+ students to understand **MVC architecture in Java**
- Hosted office hours for 6hrs/week to help students debug Java code and ask technical questions
- Graded 100+ student exams and assignments focusing on **OOP principles and software design** such as inheritance, polymorphism, encapsulation, and abstraction

Xylem Inc. Raleigh, NC
Data Science Intern May 2022—August 2022

- Developed **3 new web application screens** for a Xylem Data Lake product to drive revenue growth
- Integrated customer data with an Angular based application using **SQL and AWS**
- Developed Python scripts to analyze security concerns for **1K+ Xylem employees**
- Worked in an **Agile** environment, actively contributed to sprint planning and daily stand ups

Arduino MIDI Light Show
Personal Project July 2025

- Engineered a real-time, event-driven system integrating Python, multithreading, serial I/O, and embedded C++ to synchronize LED hardware behavior with audio playback
- Implemented a custom serial communication protocol using pyserial to transmit low-latency lighting commands to an Arduino microcontroller
- Leveraged pretty_midi and pygame to concurrently parse MIDI data and play synchronized WAV audio, enabling responsive, deterministic system behavior
- Designed the system for extensibility, allowing new lighting patterns to be mapped dynamically to musical attributes